

Lab 2: Linguistics Data

1. Introduction

This analysis examines regional variation in language use across the United States. In particular, I am interested in how linguistic responses differ across geographic areas and whether people can be grouped into meaningful clusters based on similarities in their language use.

Identifying such clusters may reveal groups of regions that share similar linguistic patterns even when they are not geographically adjacent. If comparable data were available over time, it would also be interesting to examine how these linguistic groups change. However, the current dataset does not provide this temporal dimension.

In this analysis, I first explore the geographic patterns of selected linguistic responses. I then use dimension reduction and clustering methods to examine whether respondents with similar patterns of word use can be divided into distinct groups, how many groups are supported by the data, and how stable these groupings are.

2. The Data

The data come from Bert Vaux's Dialect Survey and contain responses from 47,471 respondents across the United States. The individual-level dataset contains respondent ID, self-reported city and state, ZIP code, latitude and longitude, and responses to a set of dialect survey questions.

This analysis focuses on the lexical questions numbered Q50–Q121. The survey responses are categorical. A value of 0 indicates that the respondent did not answer the question, while the other numerical values correspond to different response choices.

Respondents from different geographic areas answered the same set of questions about their language use. These responses provide information about differences in word choice and other patterns of language use across individuals. Because geographic information is available for the respondents, linguistic similarities can be compared with geographic patterns. Individuals can therefore be clustered based on similarities in their linguistic responses, and the resulting clusters can then be examined to determine whether they correspond to meaningful geographic patterns.

However, the survey does not necessarily provide a representative sample of the U.S. population, so geographic patterns should be interpreted as patterns within the observed survey respondents rather than precise population estimates.

2.1 Data Cleaning

I first selected and explored the 67 survey questions included in the dataset. Because the original responses were categorical, I converted each response category into binary indicator variables using one-hot encoding. This initially produced 535 binary variables. A response value of 0 represents a missing or unanswered question rather than an actual linguistic response. I therefore removed the 67 binary columns corresponding to response 0, leaving 468 binary linguistic features. I also examined the number of answered questions for each respondent. Although most respondents answered all 67 questions, some respondents had missing responses.

The missing responses were particularly important for dimension reduction. In the initial PCA using all respondents, the third principal component was strongly associated with the number of questions answered (correlation approximately 0.54). This suggested that part of the PCA structure reflected response completeness rather than linguistic variation.

For the main PCA and clustering analyses, I therefore restricted the data to the 40,061 respondents who answered all 67 questions. This allowed the later analyses to focus more directly on variation in linguistic responses rather than variation in missingness.

2.2 Exploratory Data Analysis

I first examined Q68 and Q69, which ask respondents what terms they use for their maternal and paternal grandmothers. I selected these two questions because they measure closely related linguistic choices, allowing me to examine both the relationship between responses and their geographic variation.

The responses showed noticeable geographic variation. “Granny” was relatively more common in Southern and Southeastern states, including Tennessee, Alabama, and Kentucky, whereas “nana” was relatively more common in the Northeast, particularly in New England. Similar geographic patterns appeared for both maternal and paternal grandmother terminology.

The two questions were also strongly related. Among respondents who answered both questions, 55.8% used the same response label for their maternal and paternal grandmothers. For example, 72.4% of respondents who used “grandma” for their maternal grandmother also used “grandma” for their paternal grandmother. At the state level, maternal and paternal usage rates were highly correlated for “granny” ($r = 0.897$) and “nana” ($r = 0.908$).

I extended the analysis to Q70 and Q71, which ask about maternal and paternal grandfather terminology. Similar geographic patterns appeared in these questions. After harmonizing slightly different response categories between Q70 and Q71, 65.4% of respondents used the same term for both grandfathers. State-level usage was also strongly correlated for several terms, including “grandpa”

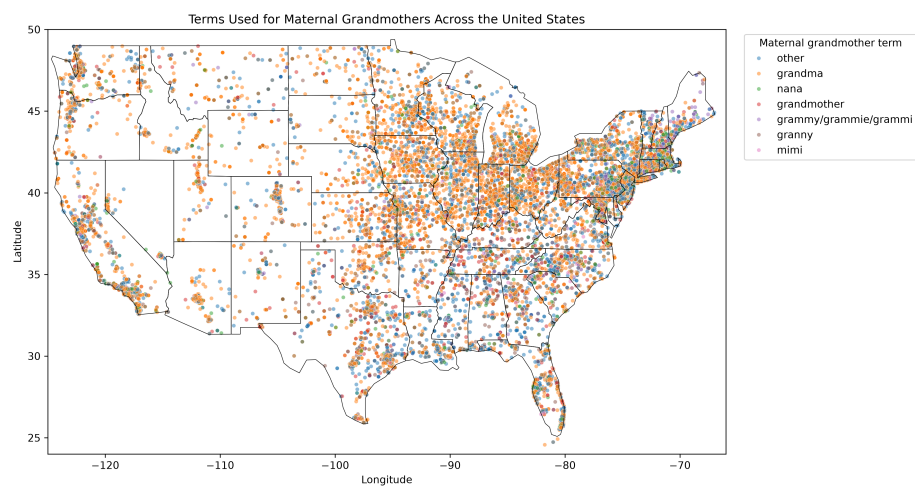


Figure 1: Geographic distribution of terms used for maternal grandmothers across the United States.

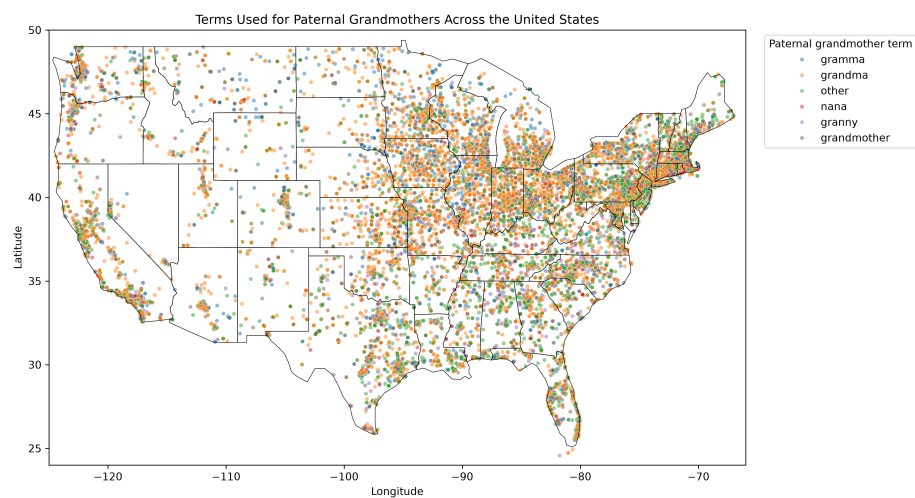


Figure 2: Geographic distribution of terms used for paternal grandmothers across the United States.

($r = 0.879$) and “grampa” ($r = 0.898$). Together, these results suggest that regional patterns in grandparent terminology are present across multiple related linguistic questions.

3. Dimension Reduction

As described in the data-cleaning section, I transformed the original categorical responses into binary variables. The initial one-hot encoding produced 535 binary variables, and after removing the 67 indicators corresponding to non-responses, 468 binary linguistic features were retained. I first performed PCA using all 47,471 respondents. The first principal component showed a noticeable geographic pattern, while the second principal component also showed some geographic structure. However, PC3 appeared to capture missingness rather than purely linguistic variation: its correlation with the number of questions answered was approximately 0.54. I therefore restricted the main PCA analysis to the 40,061 respondents who answered all 67 questions and repeated the analysis.

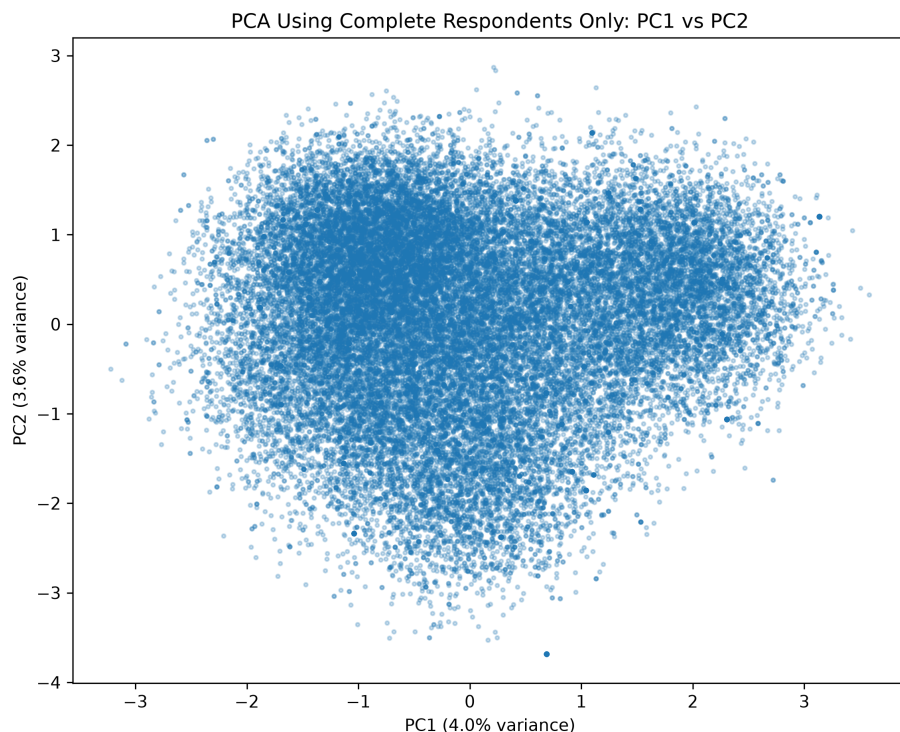


Figure 3: Projection of complete respondents onto the first two principal components.

The PC1–PC2 projection does not show clearly separated groups of respondents. Instead, the observations substantially overlap, suggesting that the linguistic

differences represented by the first two principal components may be more continuous than discrete. The first two components explain only about 7.6% of the total variance, so this two-dimensional projection captures only a limited portion of the overall variation in the data.

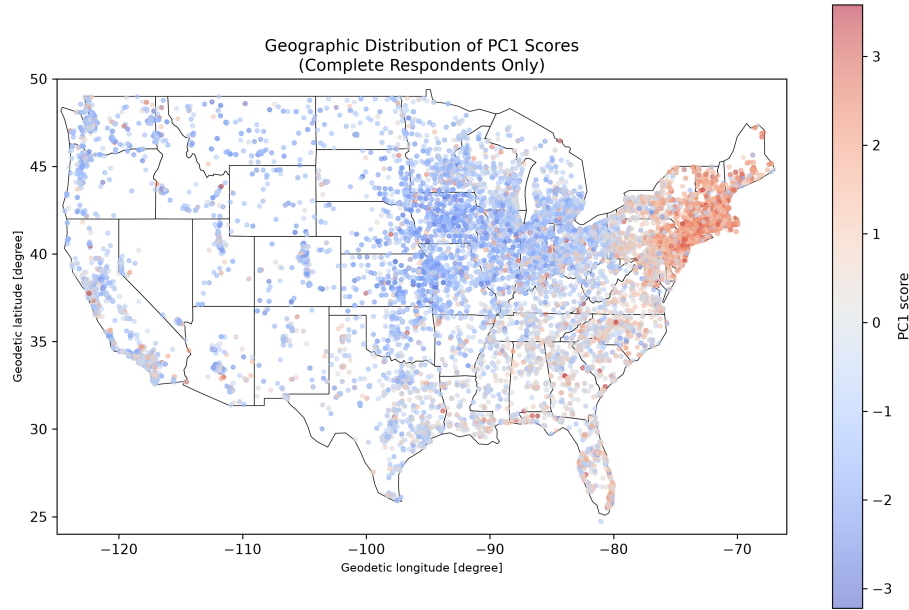


Figure 4: Geographic distribution of PC1 scores among complete respondents.

Although the PCA scatter plot does not reveal clearly separated groups, mapping the principal component scores reveals geographic structure. PC1 in particular shows a noticeable contrast between the Northeast and much of the interior and western United States. PC2 also exhibits geographic variation, although its pattern is less pronounced. Thus, the PCA suggests that some of the variation in responses is geographically structured even though respondents do not form clearly separated groups in the low-dimensional projection.

The binary features were centered but not standardized. Scikit-learn’s PCA centers each feature by default. I did not scale the features to unit variance because all features were binary indicators on the same 0–1 scale. Standardizing these indicators would give relatively greater weight to rare response categories, which I did not want to emphasize disproportionately in the PCA.

I did not apply PCA directly to the original categorical response codes because the numerical codes are arbitrary category labels rather than quantitative measurements. Treating them as numerical values would impose artificial ordering and distances between response categories. Binary encoding avoids imposing this numerical structure.

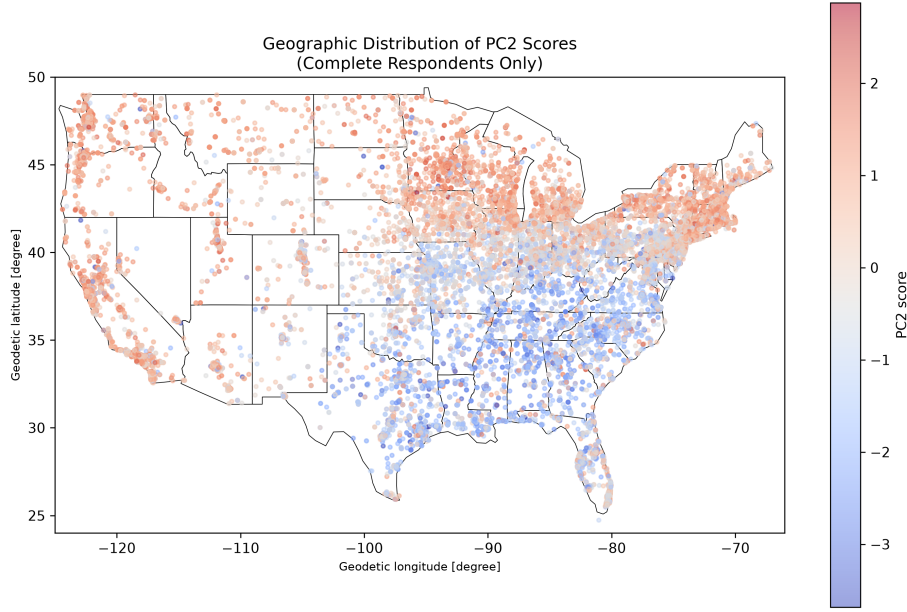


Figure 5: Geographic distribution of PC2 scores among complete respondents.

4. Clustering

I used two clustering methods, K-means and spectral clustering. For K-means, I analyzed the 40,061 respondents who answered all 67 survey questions using the 468 binary linguistic features described above. I compared solutions from $k=2$ to $k=10$. The elbow plot did not show a clear cutoff, but the silhouette score was highest at $k=3$, with a value of approximately 0.032. Based on this result, I selected three clusters for the main K-means analysis. The three clusters contained 11,433, 11,406, and 17,222 respondents, respectively. However, the silhouette score was very low even at its maximum, suggesting substantial overlap between the clusters. Therefore, the three clusters should not be interpreted as three clearly separated or objectively distinct linguistic populations.

Despite the substantial overlap, the K-means clusters showed clear geographic structure when mapped across the United States. The three clusters roughly corresponded to the Northeast, the South/interior, and the North/West/Upper Midwest, although there was considerable geographic mixing near their boundaries. Several linguistic responses also strongly distinguished the clusters. For example, “sneakers” was used by 88.8% of respondents in Cluster 1, compared with only about 15–17% in the other two clusters. “Y’all” was much more common in Cluster 0 (51.1%), while “pop” was most common in Cluster 2 (52.0%). In contrast, “soda” was especially common in Cluster 1 (84.3%). Other differences, such as “kitty-corner” versus “catty-corner” and “drinking fountain” ver-

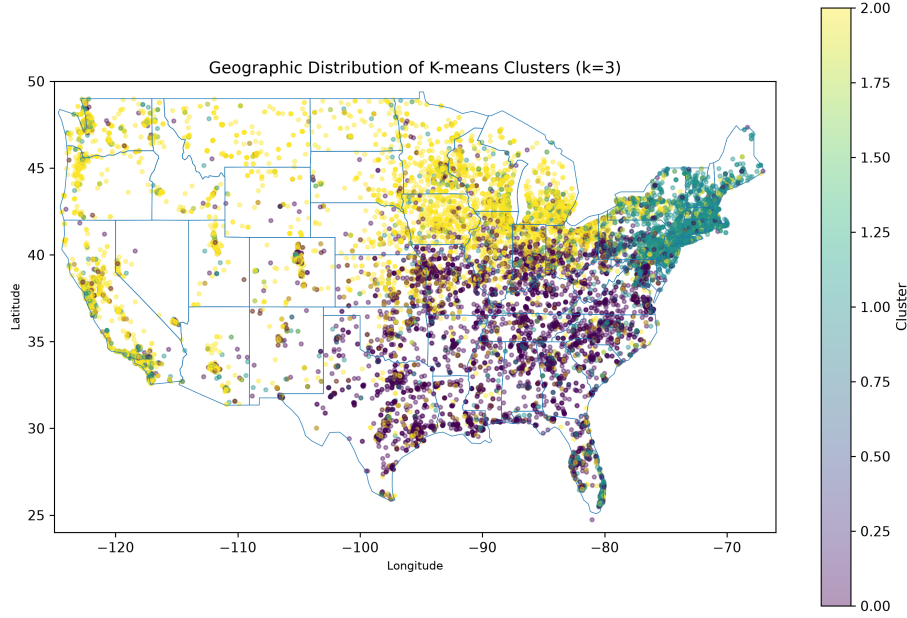


Figure 6: Geographic distribution of the three K-means clusters among complete respondents.

sus “water fountain,” also contributed to the separation of the clusters. These patterns suggest that the clusters capture meaningful regional differences in lexical usage, even though their boundaries are not sharply separated.

For spectral clustering, I used a random sample of 10,000 complete respondents because applying the method to the full dataset was computationally expensive. I performed the analysis using the first 10 PCA coordinates. With $n_neighbors=10$, the similarity graph was disconnected and produced an uninformative clustering result. I therefore examined graph connectivity using different values of $n_neighbors$. The graph became fully connected at $n_neighbors=200$, so I used this value for the final spectral clustering analysis.

I used $k=3$ for spectral clustering primarily to make the results directly comparable with the three-cluster K-means solution. The eigenvalue gaps did not provide strong evidence for exactly three spectral clusters, so $k=3$ should not be interpreted as a uniquely supported number of clusters for this method.

The spectral clustering result differed substantially from the K-means solution. One spectral cluster captured the Northeast in a way that was quite similar to the corresponding K-means cluster. However, spectral clustering combined most of the remaining respondents into one large cluster and separated a much smaller cluster containing only 177 respondents. The overall agreement between the two methods was moderate, with an Adjusted Rand Index (ARI) of approx-

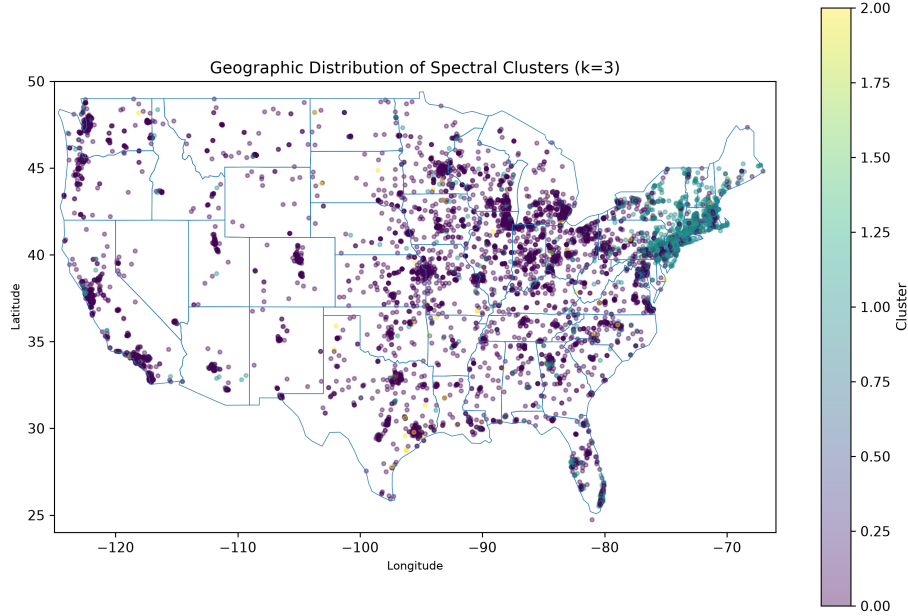


Figure 7: Geographic distribution of the three spectral clusters in the 10,000-respondent subsample.

imately 0.387 and a Normalized Mutual Information (NMI) of approximately 0.510. Thus, the two methods identified some common geographic structure, particularly in the Northeast, but did not produce the same overall partition of respondents.

The two methods also have different advantages and limitations for these data. K-means is computationally efficient and produces clusters that are relatively easy to interpret, but it relies on Euclidean distance and centroid-based separation, which may not fully represent the structure of binary linguistic features. Spectral clustering can capture more complex, nonlinear structures through a similarity graph, but it is substantially more computationally expensive and, in this analysis, was sensitive to the choice of the number of neighbors. Overall, the geographic patterns and differences in lexical responses provide evidence of regional linguistic structure, but the low silhouette scores and the differences between K-means and spectral clustering suggest that this structure is not composed of sharply separated groups. Instead, the results are more consistent with overlapping or continuous linguistic variation across regions.

5. Stability of Findings to Perturbation

I evaluated the stability of the K-means clustering results in two ways. First, I repeated K-means using different random initializations. The resulting cluster

assignments were nearly identical to the original solution, with ARI values ranging from approximately 0.991 to 0.997. This suggests that the K-means result is not sensitive to random initialization.

Second, I repeatedly selected random 80% subsamples of the complete respondents and compared their cluster assignments with those from the full-sample K-means solution. Across 10 repetitions, the ARI ranged from approximately 0.984 to 0.993, with a mean of 0.988. This indicates that the K-means solution is also highly stable to moderate changes in the sample.

However, this stability does not imply that three clusters are the uniquely correct representation of the data. As discussed above, the low silhouette score and the differences between K-means and spectral clustering suggest that the underlying linguistic variation may still be overlapping and continuous.

6. Conclusion

This analysis shows that lexical variation in the United States has meaningful geographic structure, although the boundaries between linguistic groups are not clearly discrete. PCA revealed geographic patterns while also showing substantial overlap among respondents, and K-means identified three geographically interpretable clusters. However, the low silhouette score and differences between K-means and spectral clustering suggest that linguistic variation is better understood as overlapping patterns rather than three clearly separated populations.

These patterns may be useful for future linguistic research and language technologies by documenting how word choices differ across regions and individuals. As a reality check, the geographic patterns identified here could be compared with independently documented patterns of American regional dialects. With more time and computational resources, I would also apply spectral clustering to the full dataset rather than the 10,000-person subsample. Additional demographic information, such as age, race or ethnicity, gender, and income, could also help examine how linguistic variation is associated with individual characteristics beyond geography.

7. Academic Honesty

7.1 Statement

I declare that this report represents my original work completed for this course. All data processing, model implementation, and written interpretations were conducted in accordance with the university and course academic integrity policies. Where external research and datasets were utilized, they have been properly credited and cited.

As researchers, it is important that we remain truthful about our work, since a lack of honesty can inhibit collaborative progress. Maintaining academic in-

tegrity is essential for producing reliable and reproducible research.

7.2 LLM Usage

I used ChatGPT to help revise grammar, wording, translating, and clarity in sections of the report based on my existing drafts and interpretations. I also used ChatGPT to help revise and debug portions of my existing code and to clarify the interpretation of some analytical results. I reviewed and edited the suggestions before incorporating them into my analysis and report.

7.3 Collaborators

I did not collaborate with other students on this lab.

8 Bibliography

Nerbonne, J., & Kretzschmar, W. (2003). Introducing computational techniques in dialectometry. *Computers and the Humanities*, 37(3), 245–255.

Nerbonne, J., & Kretzschmar, W. (2006). Progress in dialectometry: Toward explanation. *Literary and Linguistic Computing*, 21(4), 387–397.